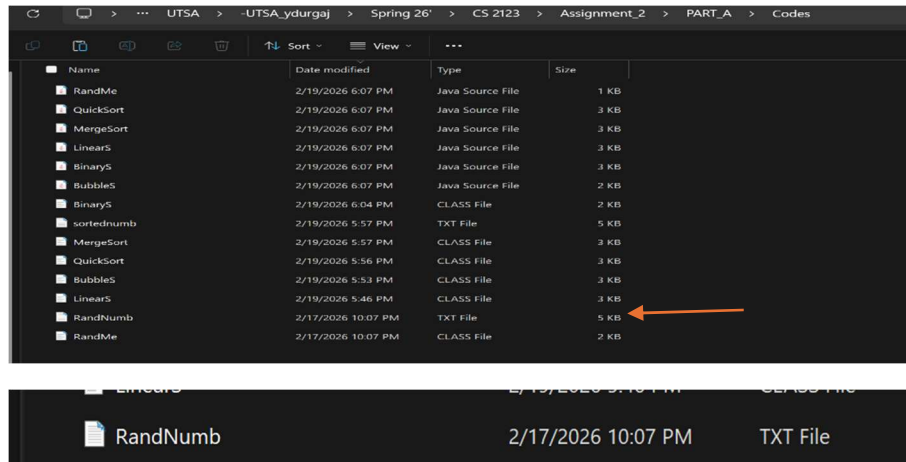


Assignment – 2 (Part 1 - Screen shots)

- Convert to PDF before submission on canvas
1. Random number generated file screenshot
 - a. Screenshot the directory with file. For example:



2. CPU time taken by linear search for each number
 - a. Output screenshot
 - i. <Image> “Student or Group names and ABC123. Linear Search # 1 search time: <time in sec>”

```
PS D:\UTSA\~\UTSA_ydurgaj\Spring 26'\CS 2123\Assignment 2\PART_A> javac LinearS.java
PS D:\UTSA\~\UTSA_ydurgaj\Spring 26'\CS 2123\Assignment 2\PART_A> java LinearS.java
Element 1 found at index: 918
Jasen Daniel Drileck (VHA667), Durga Jhansi Yerukala (SCP203), Vyshanavi Sai Yadlapati (FPL006).
Linear Search # 1 search time: 0.062500000000 sec
```

- b. Output screenshot
 - i. <Image> “Student or Group names and ABC123. Linear Search # 500 search time: <time in sec>”

```
PS D:\UTSA\~\UTSA_ydurgaj\Spring 26'\CS 2123\Assignment 2\PART_A> javac LinearS.java
PS D:\UTSA\~\UTSA_ydurgaj\Spring 26'\CS 2123\Assignment 2\PART_A> java LinearS.java
Element 500 found at index: 110
Jasen Daniel Drileck (VHA667), Durga Jhansi Yerukala (SCP203), Vyshanavi Sai Yadlapati (FPL006).
Linear Search # 500 search time: 0.046875000000 sec
```

- c. Output screenshot
 - i. <Image> “Student or Group names and ABC123. Linear Search # 900 search time: <time in sec>”

```
PS D:\UTSA\~\UTSA_ydurgaj\Spring 26'\CS 2123\Assignment 2\PART_A> javac LinearS.java
PS D:\UTSA\~\UTSA_ydurgaj\Spring 26'\CS 2123\Assignment 2\PART_A> java LinearS.java
Element 900 found at index: 760
Jasen Daniel Drileck (VHA667), Durga Jhansi Yerukala (SCP203), Vyshanavi Sai Yadlapati (FPL006).
Linear Search # 900 search time: 0.078125000000 sec
```

3. CPU time taken by different sorting algorithms

a. Output screenshot

- i. <Image> “Student or Group names and ABC123. Bubble Sort time: <time in sec>”

```
PS D:\UTSA\UTSA_ydurgaj\Spring 26'\CS 2123\Assignment_2\PART_A> javac BubbleS.java
PS D:\UTSA\UTSA_ydurgaj\Spring 26'\CS 2123\Assignment_2\PART_A> java BubbleS.java
Array sorted using Bubble Sort
Jasen Daniel Drileck (VHA667), Durga Jhansi Yerukala (SCP203), Vyshanavi Sai Yadlapati (FPL006).
Bubble Sort time: 0.046875 sec
```

b. Output screenshot

- i. <Image> “Student or Group names and ABC123. Merge Sort time: <time in sec>”

```
PS D:\UTSA\UTSA_ydurgaj\Spring 26'\CS 2123\Assignment_2\PART_A> javac MergeSort.java
PS D:\UTSA\UTSA_ydurgaj\Spring 26'\CS 2123\Assignment_2\PART_A> java MergeSort.java
Array sorted using Merge Sort
Jasen Daniel Drileck (VHA667), Durga Jhansi Yerukala (SCP203), Vyshanavi Sai Yadlapati (FPL006).
Merge Sort time: 0.0625 sec
```

c. Output screenshot

- i. <Image> “Student or Group names and ABC123. Quick Sort time: <time in sec>”

```
PS D:\UTSA\UTSA_ydurgaj\Spring 26'\CS 2123\Assignment_2\PART_A> javac QuickSort.java
PS D:\UTSA\UTSA_ydurgaj\Spring 26'\CS 2123\Assignment_2\PART_A> java QuickSort.java
Array sorted using Quick Sort (Pivot = First Index)
Jasen Daniel Drileck (VHA667), Durga Jhansi Yerukala (SCP203), Vyshanavi Sai Yadlapati (FPL006).
Quick Sort (Pivot = First Index) time: 0.062500 sec
```

4. CPU time taken by Binary search for each number

a. Output screenshot

- i. <Image> “Student or Group names and ABC123 Binary Search # 1 search time: <time in sec>”

```
PS D:\UTSA\UTSA_ydurgaj\Spring 26'\CS 2123\Assignment_2\PART_A> javac BinaryS.java
PS D:\UTSA\UTSA_ydurgaj\Spring 26'\CS 2123\Assignment_2\PART_A> java BinaryS.java
1 found at index: 1
Jasen Daniel Drileck (VHA667), Durga Jhansi Yerukala (SCP203), Vyshanavi Sai Yadlapati (FPL006).
Binary Search # 1 search time: 0.046875 sec
```

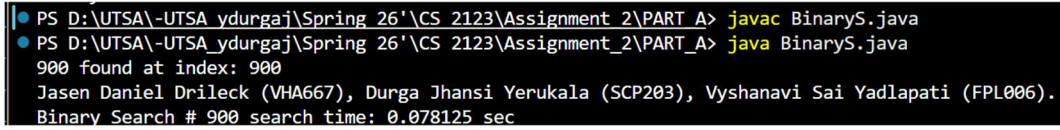
b. Output screenshot

- i. <Image> “Student or Group names and ABC123 Binary Search # 500 search time: <time in sec>”

```
PS D:\UTSA\UTSA_ydurgaj\Spring 26'\CS 2123\Assignment_2\PART_A> javac BinaryS.java
PS D:\UTSA\UTSA_ydurgaj\Spring 26'\CS 2123\Assignment_2\PART_A> java BinaryS.java
500 found at index: 500
Jasen Daniel Drileck (VHA667), Durga Jhansi Yerukala (SCP203), Vyshanavi Sai Yadlapati (FPL006).
Binary Search # 500 search time: 0.046875 sec
```

c. Output screenshot

- i. <Image> “Student or Group names and ABC123 Binary Search # 900 search time: <time in sec>”



```
PS D:\UTSA\~\UTSA_ydurgaj\Spring 26'\CS 2123\Assignment 2\PART_A> javac BinaryS.java
PS D:\UTSA\~\UTSA_ydurgaj\Spring 26'\CS 2123\Assignment 2\PART_A> java BinaryS.java
900 found at index: 900
Jasen Daniel Drileck (VHA667), Durga Jhansi Yerukala (SCP203), Vyshanavi Sai Yadlapati (FPL006).
Binary Search # 900 search time: 0.078125 sec
```